# GISviz

April 13, 2026

[1]:
```python
from pathlib import Path
import os
import json
import math
import glob

import numpy as np
import pandas as pd

try:
    import geopandas as gpd
except ImportError:
    gpd = None

try:
    import rasterio
    from rasterio.transform import from_origin
except ImportError:
    rasterio = None

try:
    from scipy.interpolate import griddata
except ImportError:
    griddata = None
```

[2]:
```python
# --- main project folders ---
ROOT = Path("/home/jovyan")
RAW_DIR = ROOT / "raw_data"
DEPTH_GRID_DIR = RAW_DIR / "north_slope_depth_grids"

# where your parquet layers are / will be
PARQUET_DIRS = [
    ROOT / "parquets",
    ROOT / "processed_data",
    ROOT / "raw_data",
]

# output folder for rasters made from XYZ surfaces
```

```python
RASTER_OUT_DIR = DEPTH_GRID_DIR / "rasters_from_xyz"
RASTER_OUT_DIR.mkdir(parents=True, exist_ok=True)

# manifest outputs
MANIFEST_DIR = ROOT / "project_manifests"
MANIFEST_DIR.mkdir(parents=True, exist_ok=True)

TXT_MANIFEST = MANIFEST_DIR / "north_slope_file_manifest.txt"
CSV_MANIFEST = MANIFEST_DIR / "north_slope_file_manifest.csv"

print("DEPTH_GRID_DIR:", DEPTH_GRID_DIR)
print("RASTER_OUT_DIR:", RASTER_OUT_DIR)
print("MANIFEST_DIR:", MANIFEST_DIR)
```

```
DEPTH_GRID_DIR: /home/jovyan/raw_data/north_slope_depth_grids
RASTER_OUT_DIR: /home/jovyan/raw_data/north_slope_depth_grids/rasters_from_xyz
MANIFEST_DIR: /home/jovyan/project_manifests
```

[3]:
```python
def scan_files(base_dirs, extensions=None):
    rows = []

    for base in base_dirs:
        if not base.exists():
            continue

        for path in base.rglob("*"):
            if path.is_file():
                if extensions:
                    if path.suffix.lower() not in extensions:
                        continue

                rows.append({
                    "name": path.name,
                    "suffix": path.suffix.lower(),
                    "parent": str(path.parent),
                    "full_path": str(path.resolve()),
                    "size_mb": round(path.stat().st_size / (1024**2), 3),
                    "modified": pd.to_datetime(path.stat().st_mtime, unit="s"),
                })

    df = pd.DataFrame(rows).sort_values(["suffix", "name", "modified"]).
 ↪reset_index(drop=True)
    return df

base_dirs_to_scan = [
    ROOT,
]
```

```python
manifest_df = scan_files(
    base_dirs_to_scan,
    extensions={".parquet", ".xyz", ".tif", ".tiff", ".ipynb", ".csv", ".txt",␣
 ↪".shp", ".geojson"}
)

manifest_df.head(20)
print(f"Found {len(manifest_df)} matching files.")
```

```
Found 804 matching files.
```

[4]:
```python
manifest_df.to_csv(CSV_MANIFEST, index=False)

with open(TXT_MANIFEST, "w", encoding="utf-8") as f:
    f.write("NORTH SLOPE PROJECT FILE MANIFEST\n")
    f.write("=" * 80 + "\n\n")

    for _, row in manifest_df.iterrows():
        f.write(f"NAME:     {row['name']}\n")
        f.write(f"TYPE:     {row['suffix']}\n")
        f.write(f"SIZE_MB:  {row['size_mb']}\n")
        f.write(f"MODIFIED: {row['modified']}\n")
        f.write(f"PARENT:   {row['parent']}\n")
        f.write(f"PATH:     {row['full_path']}\n")
        f.write("-" * 80 + "\n")

print("Saved:")
print(TXT_MANIFEST)
print(CSV_MANIFEST)
```

```
Saved:
/home/jovyan/project_manifests/north_slope_file_manifest.txt
/home/jovyan/project_manifests/north_slope_file_manifest.csv
```

[5]:
```python
parquet_df = manifest_df[manifest_df["suffix"] == ".parquet"].copy()
display(parquet_df[["name", "parent", "full_path", "size_mb"]].
 ↪sort_values("name"))
print(f"Total parquet files: {len(parquet_df)}")
```

```
                                             name  \
330                         alaska_base_map.parquet
331                           clean_2d_seismic.parquet
332                  clean_2d_seismic_features.parquet
333                           clean_3d_seismic.parquet
334                  clean_3d_seismic_features.parquet
335                       clean_well_locations.parquet
336              clean_well_locations_features.parquet
337        clean_well_locations_features_dedup.parquet
```

```
338                north_slope_assessment_units.parquet
339       north_slope_assessment_units_features.parquet
340                          north_slope_extent.parquet
341                 north_slope_extent_features.parquet
342                north_slope_master_2d_layers.parquet
343              north_slope_master_3d_surfaces.parquet
344             north_slope_structural_surfaces.parquet
345    north_slope_structural_surfaces_3d_ready.parquet
346   north_slope_structural_surfaces_3d_ready_dedup…
347   north_slope_structural_surfaces_in_3d_extent.p…

                                              parent  \
330    /home/jovyan/notebooks/03_data_final/core_layers
331    /home/jovyan/notebooks/03_data_final/core_layers
332   /home/jovyan/notebooks/03_data_final/feature_l…
333    /home/jovyan/notebooks/03_data_final/core_layers
334   /home/jovyan/notebooks/03_data_final/feature_l…
335    /home/jovyan/notebooks/03_data_final/core_layers
336   /home/jovyan/notebooks/03_data_final/feature_l…
337   /home/jovyan/notebooks/03_data_final/feature_l…
338    /home/jovyan/notebooks/03_data_final/core_layers
339   /home/jovyan/notebooks/03_data_final/feature_l…
340    /home/jovyan/notebooks/03_data_final/core_layers
341   /home/jovyan/notebooks/03_data_final/feature_l…
342   /home/jovyan/notebooks/03_data_final/master_la…
343   /home/jovyan/notebooks/03_data_final/master_la…
344   /home/jovyan/notebooks/04_data_intermediate/st…
345   /home/jovyan/notebooks/04_data_intermediate/st…
346   /home/jovyan/notebooks/04_data_intermediate/st…
347   /home/jovyan/notebooks/04_data_intermediate/st…

                                           full_path  size_mb
330   /home/jovyan/notebooks/03_data_final/core_laye…    0.101
331   /home/jovyan/notebooks/03_data_final/core_laye…    4.930
332   /home/jovyan/notebooks/03_data_final/feature_l…    2.884
333   /home/jovyan/notebooks/03_data_final/core_laye…    1.948
334   /home/jovyan/notebooks/03_data_final/feature_l…    1.552
335   /home/jovyan/notebooks/03_data_final/core_laye…    0.192
336   /home/jovyan/notebooks/03_data_final/feature_l…    0.966
337   /home/jovyan/notebooks/03_data_final/feature_l…    0.819
338   /home/jovyan/notebooks/03_data_final/core_laye…    1.034
339   /home/jovyan/notebooks/03_data_final/feature_l…    1.133
340   /home/jovyan/notebooks/03_data_final/core_laye…    0.010
341   /home/jovyan/notebooks/03_data_final/feature_l…    0.008
342   /home/jovyan/notebooks/03_data_final/master_la…   18.066
343   /home/jovyan/notebooks/03_data_final/master_la…    6.414
344   /home/jovyan/notebooks/04_data_intermediate/st…    9.645
345   /home/jovyan/notebooks/04_data_intermediate/st…    0.142
```

```
346  /home/jovyan/notebooks/04_data_intermediate/st…    0.092
347  /home/jovyan/notebooks/04_data_intermediate/st…    0.106

Total parquet files: 18
```

[6]:

```python
loaded_layers = {}

for _, row in parquet_df.iterrows():
    path = Path(row["full_path"])
    key = path.stem.replace(" ", "_").replace("-", "_")

    try:
        df = pd.read_parquet(path)
        loaded_layers[key] = df
        print(f"Loaded: {key:40s} | shape={df.shape}")
    except Exception as e:
        print(f"FAILED: {path.name} -> {e}")

print(f"\nLoaded {len(loaded_layers)} parquet datasets.")
```

```
Loaded: alaska_base_map                          | shape=(9, 169)
Loaded: clean_2d_seismic                         | shape=(26, 7)
Loaded: clean_2d_seismic_features                | shape=(8, 13)
Loaded: clean_3d_seismic                         | shape=(36, 8)
Loaded: clean_3d_seismic_features                | shape=(24, 15)
Loaded: clean_well_locations                     | shape=(10250, 4)
Loaded: clean_well_locations_features            | shape=(33078, 23)
Loaded: clean_well_locations_features_dedup      | shape=(9861, 24)
Loaded: north_slope_assessment_units             | shape=(6, 6)
Loaded: north_slope_assessment_units_features    | shape=(6, 10)
Loaded: north_slope_extent                       | shape=(1, 2)
Loaded: north_slope_extent_features              | shape=(1, 2)
Loaded: north_slope_master_2d_layers             | shape=(538877, 26)
Loaded: north_slope_master_3d_surfaces           | shape=(723840, 8)
Loaded: north_slope_structural_surfaces          | shape=(723840, 9)
Loaded: north_slope_structural_surfaces_3d_ready | shape=(13416, 12)
Loaded: north_slope_structural_surfaces_3d_ready_dedup | shape=(7380, 12)
Loaded: north_slope_structural_surfaces_in_3d_extent | shape=(9840, 9)

Loaded 18 parquet datasets.
```

[7]:

```python
for name, df in loaded_layers.items():
    print("\n" + "=" * 80)
    print(name)
    print(df.head(3))
    print(df.columns.tolist())
```

```
================================================================================
```

```
alaska_base_map
        featurecla  scalerank  LABELRANK                SOVEREIGNT SOV_A3  \
4  Admin-0 country          1          2  United States of America    US1
4  Admin-0 country          1          2  United States of America    US1
4  Admin-0 country          1          2  United States of America    US1

   ADM0_DIF  LEVEL     TYPE TLC                     ADMIN  … FCLASS_TR  \
4         1      2  Country   1  United States of America  …       NaN
4         1      2  Country   1  United States of America  …       NaN
4         1      2  Country   1  United States of America  …       NaN

   FCLASS_ID FCLASS_PL FCLASS_GR  FCLASS_IT FCLASS_NL FCLASS_SE  FCLASS_BD  \
4        NaN       NaN       NaN        NaN       NaN       NaN        NaN
4        NaN       NaN       NaN        NaN       NaN       NaN        NaN
4        NaN       NaN       NaN        NaN       NaN       NaN        NaN

  FCLASS_UA                                          geometry
4       NaN  b"\x01\x03\x00\x00\x00\x01\x00\x00\x00\x11\x00…
4       NaN  b"\x01\x03\x00\x00\x00\x01\x00\x00\x00\t\x00\x…
4       NaN  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\x05\x00…

[3 rows x 169 columns]
['featurecla', 'scalerank', 'LABELRANK', 'SOVEREIGNT', 'SOV_A3', 'ADM0_DIF',
'LEVEL', 'TYPE', 'TLC', 'ADMIN', 'ADM0_A3', 'GEOU_DIF', 'GEOUNIT', 'GU_A3',
'SU_DIF', 'SUBUNIT', 'SU_A3', 'BRK_DIFF', 'NAME', 'NAME_LONG', 'BRK_A3',
'BRK_NAME', 'BRK_GROUP', 'ABBREV', 'POSTAL', 'FORMAL_EN', 'FORMAL_FR',
'NAME_CIAWF', 'NOTE_ADM0', 'NOTE_BRK', 'NAME_SORT', 'NAME_ALT', 'MAPCOLOR7',
'MAPCOLOR8', 'MAPCOLOR9', 'MAPCOLOR13', 'POP_EST', 'POP_RANK', 'POP_YEAR',
'GDP_MD', 'GDP_YEAR', 'ECONOMY', 'INCOME_GRP', 'FIPS_10', 'ISO_A2', 'ISO_A2_EH',
'ISO_A3', 'ISO_A3_EH', 'ISO_N3', 'ISO_N3_EH', 'UN_A3', 'WB_A2', 'WB_A3',
'WOE_ID', 'WOE_ID_EH', 'WOE_NOTE', 'ADM0_ISO', 'ADM0_DIFF', 'ADM0_TLC',
'ADM0_A3_US', 'ADM0_A3_FR', 'ADM0_A3_RU', 'ADM0_A3_ES', 'ADM0_A3_CN',
'ADM0_A3_TW', 'ADM0_A3_IN', 'ADM0_A3_NP', 'ADM0_A3_PK', 'ADM0_A3_DE',
'ADM0_A3_GB', 'ADM0_A3_BR', 'ADM0_A3_IL', 'ADM0_A3_PS', 'ADM0_A3_SA',
'ADM0_A3_EG', 'ADM0_A3_MA', 'ADM0_A3_PT', 'ADM0_A3_AR', 'ADM0_A3_JP',
'ADM0_A3_KO', 'ADM0_A3_VN', 'ADM0_A3_TR', 'ADM0_A3_ID', 'ADM0_A3_PL',
'ADM0_A3_GR', 'ADM0_A3_IT', 'ADM0_A3_NL', 'ADM0_A3_SE', 'ADM0_A3_BD',
'ADM0_A3_UA', 'ADM0_A3_UN', 'ADM0_A3_WB', 'CONTINENT', 'REGION_UN', 'SUBREGION',
'REGION_WB', 'NAME_LEN', 'LONG_LEN', 'ABBREV_LEN', 'TINY', 'HOMEPART',
'MIN_ZOOM', 'MIN_LABEL', 'MAX_LABEL', 'LABEL_X', 'LABEL_Y', 'NE_ID',
'WIKIDATAID', 'NAME_AR', 'NAME_BN', 'NAME_DE', 'NAME_EN', 'NAME_ES', 'NAME_FA',
'NAME_FR', 'NAME_EL', 'NAME_HE', 'NAME_HI', 'NAME_HU', 'NAME_ID', 'NAME_IT',
'NAME_JA', 'NAME_KO', 'NAME_NL', 'NAME_PL', 'NAME_PT', 'NAME_RU', 'NAME_SV',
'NAME_TR', 'NAME_UK', 'NAME_UR', 'NAME_VI', 'NAME_ZH', 'NAME_ZHT', 'FCLASS_ISO',
'TLC_DIFF', 'FCLASS_TLC', 'FCLASS_US', 'FCLASS_FR', 'FCLASS_RU', 'FCLASS_ES',
'FCLASS_CN', 'FCLASS_TW', 'FCLASS_IN', 'FCLASS_NP', 'FCLASS_PK', 'FCLASS_DE',
'FCLASS_GB', 'FCLASS_BR', 'FCLASS_IL', 'FCLASS_PS', 'FCLASS_SA', 'FCLASS_EG',
'FCLASS_MA', 'FCLASS_PT', 'FCLASS_AR', 'FCLASS_JP', 'FCLASS_KO', 'FCLASS_VN',
```

```
'FCLASS_TR', 'FCLASS_ID', 'FCLASS_PL', 'FCLASS_GR', 'FCLASS_IT', 'FCLASS_NL',
'FCLASS_SE', 'FCLASS_BD', 'FCLASS_UA', 'geometry']

================================================================================
clean_2d_seismic
                               survey_name  year_acquired  year_released  \
0  Pike Bottom Hole Planning 2D Reprocessing            NaN         2018.0
1                               Stingray 2D         2011.0         2021.0
2                    Nenana Basin 2D (2005)         2005.0         2019.0

                                             company  line_miles  \
0                                 WesternGeco and CPA      153.75
1  Northern Exploration Services LLC & Fairfield …        2.24
2              Weiman GeoScience & Geo Concepts Inc.      197.21

                                            geometry type
0  b'\x01\x05\x00\x00\x00)\x00\x00\x00\x01\x02\x0…   2D
1  b'\x01\x02\x00\x00\x00\x9e\x00\x00\x00\x05\xcb…   2D
2  b'\x01\x05\x00\x00\x00S\x00\x00\x00\x01\x02\x0…   2D
['survey_name', 'year_acquired', 'year_released', 'company', 'line_miles',
'geometry', 'type']

================================================================================
clean_2d_seismic_features
                               survey_name  year_acquired  year_released  \
0  Pike Bottom Hole Planning 2D Reprocessing            NaN         2018.0
3                            White Hills 2D         2007.0         2017.0
6             Toolik 2D (aka Foothills 2D)         2006.0         2017.0

               company  line_miles  \
0  WesternGeco and CPA      153.75
3             Paradigm      170.42
6  Thrust Belt Imaging      243.26

                                            geometry type   length_km  \
0  b"\x01\x05\x00\x00\x00)\x00\x00\x00\x01\x02\x0…   2D  178.097855
3  b'\x01\x05\x00\x00\x00\'\x00\x00\x00\x01\x02\x…   2D  273.654439
6  b'\x01\x05\x00\x00\x00:\x00\x00\x00\x01\x02\x0…   2D  391.271409

   length_mi_geom  has_company  has_release_year  has_acquired_year  \
0      110.664842         True              True              False
3      170.040933         True              True               True
6      243.124707         True              True               True

   survey_age_years
0               NaN
3              19.0
6              20.0
```

```
['survey_name', 'year_acquired', 'year_released', 'company', 'line_miles',
'geometry', 'type', 'length_km', 'length_mi_geom', 'has_company',
'has_release_year', 'has_acquired_year', 'survey_age_years']

================================================================================
clean_3d_seismic
                                         survey_name  year_acquired  \
0                      Aklaq, Smith Bay & Simpson 3D           2008
1                                        Big Bend 3D           2014
2  Big Island / North Island 3D (aka North Slope …          2008

   year_released                company  square_miles survey_type  \
0         2019.0            WesternGeco        602.04          3D
1         2024.0  Geotrace Technologies         83.71          3D
2         2019.0            CGG Veritas        199.39          3D

                                            geometry type
0  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00M\x0f\x0…   3D
1  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\x86\t\x…   3D
2  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xf1\x06…   3D
['survey_name', 'year_acquired', 'year_released', 'company', 'square_miles',
'survey_type', 'geometry', 'type']

================================================================================
clean_3d_seismic_features
                                         survey_name  year_acquired  \
0                      Aklaq, Smith Bay & Simpson 3D           2008
1                                        Big Bend 3D           2014
2  Big Island / North Island 3D (aka North Slope …          2008

   year_released                company  square_miles survey_type  \
0         2019.0            WesternGeco        602.04          3D
1         2024.0  Geotrace Technologies         83.71          3D
2         2019.0            CGG Veritas        199.39          3D

                                            geometry type     area_km2  \
0  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00M\x0f\x0…   3D  1563.538878
1  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\x86\t\x…   3D   216.850572
2  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xf1\x06…   3D   516.593739

   area_mi2_geom  has_company  has_release_year  survey_age_years  \
0     603.685488         True              True              18.0
1      83.726439         True              True              12.0
2     199.457876         True              True              18.0

   centroid_lon  centroid_lat
0   -154.634902     70.753700
1   -151.171048     69.995037
```

```
2   -150.797844     70.214099
['survey_name', 'year_acquired', 'year_released', 'company', 'square_miles',
'survey_type', 'geometry', 'type', 'area_km2', 'area_mi2_geom', 'has_company',
'has_release_year', 'survey_age_years', 'centroid_lon', 'centroid_lat']

================================================================================
clean_well_locations
                          operator         field  \
0                  Romig Park Inc.  *EXPLORATORY
1    Anchorage Oil and Development  *EXPLORATORY
2  Union Oil Company of California  *EXPLORATORY

                                            geometry  type
0  b'\x01\x01\x00\x00\x00\x80=\nW:W\nA\xc0\x86\xa…  well
1  b'\x01\x01\x00\x00\x00@EG\xf2Y\x9b\x0bA\xc0\x9…  well
2  b'\x01\x01\x00\x00\x00\x007\x1a\xc0I\x9b\nAx$\…  well
['operator', 'field', 'geometry', 'type']

================================================================================
clean_well_locations_features
                          operator         field  \
0                  Romig Park Inc.  *EXPLORATORY
1    Anchorage Oil and Development  *EXPLORATORY
2  Union Oil Company of California  *EXPLORATORY

                                            geometry  type  has_geometry  \
0  b'\x01\x01\x00\x00\x005i\xa6\xb0E\xbfb\xc0N\xe…  well          True
1  b'\x01\x01\x00\x00\x00\x9dZI\xcc\xc6\xb8b\xc0\…  well          True
2  b'\x01\x01\x00\x00\x00*\x08\xd2\xaa0\xbdb\xc0\…  well          True

   has_field  has_operator  is_exploratory  is_strat_test  in_3d  …  \
0       True          True            True          False  False  …
1       True          True            True          False  False  …
2       True          True            True          False  False  …

   near_2d_lt_1km  near_2d_lt_5km  near_2d_lt_10km  au_name  province  \
0           False           False            False      NaN       NaN
1           False           False            False      NaN       NaN
2           False           False            False      NaN       NaN

   tps_name region in_assessment_unit         lon        lat
0       NaN    NaN              False -149.977257  61.137440
1       NaN    NaN              False -149.774267  61.208060
2       NaN    NaN              False -149.912191  61.327541

[3 rows x 23 columns]
['operator', 'field', 'geometry', 'type', 'has_geometry', 'has_field',
'has_operator', 'is_exploratory', 'is_strat_test', 'in_3d', 'dist_to_2d_m',
```

```
'dist_to_2d_km', 'dist_to_2d_mi', 'near_2d_lt_1km', 'near_2d_lt_5km',
'near_2d_lt_10km', 'au_name', 'province', 'tps_name', 'region',
'in_assessment_unit', 'lon', 'lat']

================================================================================
clean_well_locations_features_dedup
                                              well_id        operator  \
0  AIX Energy LLC | *EXPLORATORY | 60.579125 | -1…  AIX Energy LLC
1  AIX Energy LLC | KENAI LOOP | 60.564372 | -151…  AIX Energy LLC
2  AIX Energy LLC | KENAI LOOP | 60.567746 | -151…  AIX Energy LLC

          field                                           geometry  type  \
0  *EXPLORATORY  b'\x01\x01\x00\x00\x00pd\xac\xe0\xa9\xe6b\xc0j…  well
1    KENAI LOOP  b'\x01\x01\x00\x00\x00\xb0\x88\xc1[6\xe7b\xc02…  well
2    KENAI LOOP  b'\x01\x01\x00\x00\x00k\r#\x96\x08\xe7b\xc0\x1…  well

   has_geometry  has_field  has_operator  is_exploratory  is_strat_test  …  \
0          True       True          True            True          False  …
1          True       True          True           False          False  …
2          True       True          True           False          False  …

   near_2d_lt_1km  near_2d_lt_5km  near_2d_lt_10km  au_name  province  \
0           False           False            False      NaN       NaN
1           False           False             True      NaN       NaN
2           False           False             True      NaN       NaN

   tps_name  region in_assessment_unit        lon        lat
0       NaN     NaN              False -151.208237  60.579125
1       NaN     NaN              False -151.225386  60.564372
2       NaN     NaN              False -151.219798  60.567746

[3 rows x 24 columns]
['well_id', 'operator', 'field', 'geometry', 'type', 'has_geometry',
'has_field', 'has_operator', 'is_exploratory', 'is_strat_test', 'in_3d',
'dist_to_2d_m', 'dist_to_2d_km', 'dist_to_2d_mi', 'near_2d_lt_1km',
'near_2d_lt_5km', 'near_2d_lt_10km', 'au_name', 'province', 'tps_name',
'region', 'in_assessment_unit', 'lon', 'lat']

================================================================================
north_slope_assessment_units
          region         province                 tps_name  \
0  North America  Northern Alaska  Arctic Alaska Composite
1  North America  Northern Alaska  Arctic Alaska Composite
2  North America  Northern Alaska  Arctic Alaska Composite

                      au_name  \
0             Brookian Topset
1  Brookian Foreset-Bottomset
```

```
2    Beaufortian Strata North

                                              geometry             type
0  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xedD\x0…  assessment_unit
1  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xdeG\x0…  assessment_unit
2  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00=B\x00\x…  assessment_unit
['region', 'province', 'tps_name', 'au_name', 'geometry', 'type']

================================================================================
north_slope_assessment_units_features
          region         province                 tps_name  \
0  North America  Northern Alaska  Arctic Alaska Composite
1  North America  Northern Alaska  Arctic Alaska Composite
2  North America  Northern Alaska  Arctic Alaska Composite

                      au_name  \
0             Brookian Topset
1  Brookian Foreset-Bottomset
2    Beaufortian Strata North

                                              geometry             type  \
0  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xedD\x0…  assessment_unit
1  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\xdeG\x0…  assessment_unit
2  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00=B\x00\x…  assessment_unit

       area_km2  area_mi2_geom  centroid_lon  centroid_lat
0  28133.753271   10862.498406   -149.156259     69.830025
1  30690.717911   11849.747567   -149.307751     69.774484
2  16446.365764    6349.974714   -148.554777     70.079162
['region', 'province', 'tps_name', 'au_name', 'geometry', 'type', 'area_km2',
'area_mi2_geom', 'centroid_lon', 'centroid_lat']

================================================================================
north_slope_extent
                     name                                           geometry
0  North Slope Study Area  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\x05\x00…
['name', 'geometry']

================================================================================
north_slope_extent_features
                     name                                           geometry
0  North Slope Study Area  b'\x01\x03\x00\x00\x00\x01\x00\x00\x00\x05\x00…
['name', 'geometry']

================================================================================
north_slope_master_2d_layers
  layer_group  layer_name  feature_id  vertex_order     feature_type  \
0     seismic  seismic_2d           0             0  MultiLineString
```

```
1     seismic  seismic_2d          0            1  MultiLineString
2     seismic  seismic_2d          0            2  MultiLineString

        x_3338        y_3338  depth_m  \
0  236731.9887  2.285481e+06      0.0
1  236332.2632  2.284195e+06      0.0
2  236731.9887  2.285481e+06      0.0

                                 survey_name  year_acquired  …  \
0  Pike Bottom Hole Planning 2D Reprocessing            NaN  …
1  Pike Bottom Hole Planning 2D Reprocessing            NaN  …
2  Pike Bottom Hole Planning 2D Reprocessing            NaN  …

               source_file square_miles  survey_type operator  field  region  \
0  clean_2d_seismic.parquet          NaN          NaN      NaN    NaN     NaN
1  clean_2d_seismic.parquet          NaN          NaN      NaN    NaN     NaN
2  clean_2d_seismic.parquet          NaN          NaN      NaN    NaN     NaN

  province  tps_name au_name name
0      NaN       NaN     NaN  NaN
1      NaN       NaN     NaN  NaN
2      NaN       NaN     NaN  NaN

[3 rows x 26 columns]
['layer_group', 'layer_name', 'feature_id', 'vertex_order', 'feature_type',
'x_3338', 'y_3338', 'depth_m', 'survey_name', 'year_acquired', 'year_released',
'company', 'line_miles', 'type', 'lon', 'lat', 'source_file', 'square_miles',
'survey_type', 'operator', 'field', 'region', 'province', 'tps_name', 'au_name',
'name']

================================================================================
north_slope_master_3d_surfaces
     x_3338     y_3338  z_native         lon        lat surface_name  \
0 -630000.0  2621000.0     -67.0 -172.658577  72.824670       NStopo
1 -627000.0  2621000.0     -75.0 -172.574298  72.832387       NStopo
2 -624000.0  2621000.0     -82.0 -172.489958  72.840071       NStopo

  source_file  depth_m
0  NStopo.XYZ     67.0
1  NStopo.XYZ     75.0
2  NStopo.XYZ     82.0
['x_3338', 'y_3338', 'z_native', 'lon', 'lat', 'surface_name', 'source_file',
'depth_m']

================================================================================
north_slope_structural_surfaces
      layer_name value_type        source_file         x          y  \
0  nslcu-shublik    unknown  NSLCU-Shublik.XYZ -630000.0  2000000.0
```

```
1  nslcu-shublik    unknown  NSLCU-Shublik.XYZ -627000.0  2000000.0
2  nslcu-shublik    unknown  NSLCU-Shublik.XYZ -624000.0  2000000.0

        lon       lat    z                                         geometry
0 -168.67649  67.32879  0.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\xe09#\x…
1 -168.60884  67.33470  0.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00p"#\xc1\…
2 -168.54116  67.34059  0.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x0b…
['layer_name', 'value_type', 'source_file', 'x', 'y', 'lon', 'lat', 'z',
'geometry']

================================================================================
north_slope_structural_surfaces_3d_ready
      layer_name value_type        source_file        x          y        lon  \
0  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  66000.0  2150000.0 -152.34894
1  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  66000.0  2150000.0 -152.34894
2  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  96000.0  2150000.0 -151.59901

        lat       z                                         geometry  \
0  69.31117  1270.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x1d…
1  69.31117  1270.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x1d…
2  69.30285  1321.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00p\xf…

   index_right                     survey_name survey_type
0           35           Umiat 3D Reprocessing          3D
1           34  Umiat 3D (2D/3D Sold as a set)   2D and 3D
2           12    Gubik 3D (aka East Umiat 3D)          3D
['layer_name', 'value_type', 'source_file', 'x', 'y', 'lon', 'lat', 'z',
'geometry', 'index_right', 'survey_name', 'survey_type']

================================================================================
north_slope_structural_surfaces_3d_ready_dedup
  layer_name value_type     source_file       x          y        lon  \
0   basement      depth  NSbasement.XYZ -45000.0  2279000.0 -155.18335
1   basement      depth  NSbasement.XYZ -42000.0  2279000.0 -155.10448
2   basement      depth  NSbasement.XYZ -39000.0  2279000.0 -155.02560

        lat       z                                         geometry  \
0  70.49041 -4956.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\xf9…
1  70.49088 -5062.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x82…
2  70.49132 -5071.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x0b…

   index_right                    survey_name survey_type
0            0  Aklaq, Smith Bay & Simpson 3D          3D
1            0  Aklaq, Smith Bay & Simpson 3D          3D
2            0  Aklaq, Smith Bay & Simpson 3D          3D
['layer_name', 'value_type', 'source_file', 'x', 'y', 'lon', 'lat', 'z',
'geometry', 'index_right', 'survey_name', 'survey_type']
```

```
================================================================================
north_slope_structural_surfaces_in_3d_extent
      layer_name value_type        source_file        x          y        lon  \
0  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  66000.0  2150000.0 -152.34894
1  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  96000.0  2150000.0 -151.59901
2  nslcu-shublik    unknown  NSLCU-Shublik.XYZ  99000.0  2150000.0 -151.52405

        lat       z                                           geometry
0  69.31117  1270.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00\x1d…
1  69.30285  1321.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x00p\xf…
2  69.30185  1315.0  b'\x01\x01\x00\x00\x00\x00\x00\x00\x00\x80+\xf…
['layer_name', 'value_type', 'source_file', 'x', 'y', 'lon', 'lat', 'z',
'geometry']
```

[8]:
```
from pathlib import Path

ROOT = Path("/home/jovyan/notebooks")
RAW_DIR = ROOT / "raw_data"
DEPTH_GRID_DIR = RAW_DIR / "north_slope_depth_grids"

print("DEPTH_GRID_DIR =", DEPTH_GRID_DIR)
print("Exists?        =", DEPTH_GRID_DIR.exists())
print("Is dir?        =", DEPTH_GRID_DIR.is_dir())

xyz_files = sorted(DEPTH_GRID_DIR.glob("*.XYZ")) + sorted(DEPTH_GRID_DIR.
 ↪glob("*.xyz"))

print("\nXYZ files found:")
for p in xyz_files:
    print(" -", p.name)

print(f"\nTotal XYZ files: {len(xyz_files)}")
```

```
DEPTH_GRID_DIR = /home/jovyan/notebooks/raw_data/north_slope_depth_grids
Exists?        = True
Is dir?        = True

XYZ files found:
 - NSLCU-Shublik.XYZ
 - NSLCU.XYZ
 - NSbasement.XYZ
 - NSshublik-basement.XYZ
 - NSshublik.XYZ
 - NStopo-LCU.XYZ
 - NStopo-basement.XYZ
 - NStopo.XYZ

Total XYZ files: 8
```

[9]:
```python
import pandas as pd

def read_xyz_file(path):
    rows = []

    with open(path, "r", encoding="utf-8", errors="ignore") as f:
        for line in f:
            line = line.strip()

            if (
                not line
                or line.startswith("/")
                or ("X" in line and "Y" in line and "Z" in line)
                or "----" in line
            ):
                continue

            parts = line.split()
            if len(parts) < 5:
                continue

            try:
                x = float(parts[0])
                y = float(parts[1])
                z = float(parts[2])
                lon = float(parts[3])
                lat = float(parts[4])
                rows.append((x, y, z, lon, lat))
            except ValueError:
                continue

    if not rows:
        raise ValueError(f"No valid rows found in {path.name}")

    return pd.DataFrame(rows, columns=["x", "y", "z", "lon", "lat"])
```

[10]:
```python
test_xyz = DEPTH_GRID_DIR / "NSshublik.XYZ"
test_df = read_xyz_file(test_xyz)

print(test_df.head())
print(test_df.describe())
print("Rows:", len(test_df))
```

```
          x          y       z        lon       lat
0 -630000.0  2000000.0  -611.0 -168.67649  67.32879
1 -627000.0  2000000.0  -791.0 -168.60884  67.33470
2 -624000.0  2000000.0 -1092.0 -168.54116  67.34059
3 -621000.0  2000000.0 -1391.0 -168.47344  67.34646
```

```
4 -618000.0  2000000.0 -1668.0 -168.40570  67.35229
                   x             y             z           lon           lat
count   90480.000000  9.048000e+04  90480.000000  90480.000000  90480.000000
mean    21000.000000  2.310500e+06  -5446.249050   -153.450434     70.530463
std    376722.137017  1.801322e+05   4674.469126      9.972431      1.646659
min   -630000.000000  2.000000e+06 -16691.000000   -172.657310     67.243160
25%   -306000.000000  2.155250e+06 -10251.000000   -162.088437     69.116290
50%    21000.000000  2.310500e+06  -4526.000000   -153.440745     70.521000
75%   348000.000000  2.465750e+06  -1417.000000   -144.811743     71.936842
max    672000.000000  2.621000e+06   2084.000000   -134.169420     73.655720
Rows: 90480
```

[11]:
```python
import numpy as np
import pandas as pd

def estimate_spacing(values):
    vals = np.sort(np.unique(values))
    diffs = np.diff(vals)
    diffs = diffs[diffs > 0]
    if len(diffs) == 0:
        return None
    return float(pd.Series(diffs).mode().iloc[0])

dx = estimate_spacing(test_df["x"].values)
dy = estimate_spacing(test_df["y"].values)

print("Estimated dx:", dx)
print("Estimated dy:", dy)
print("Unique X:", test_df["x"].nunique())
print("Unique Y:", test_df["y"].nunique())
```

```
Estimated dx: 3000.0
Estimated dy: 3000.0
Unique X: 435
Unique Y: 208
```

[12]:
```python
import rasterio
from rasterio.transform import from_origin

def xyz_to_raster_xy(df, out_tif, crs=None, nodata=-9999.0):
    x_vals = np.sort(df["x"].unique())
    y_vals = np.sort(df["y"].unique())

    dx = estimate_spacing(x_vals)
    dy = estimate_spacing(y_vals)

    if dx is None or dy is None:
        raise ValueError("Could not determine x/y spacing.")
```

```python
    print(f"Grid size will be {len(y_vals)} rows x {len(x_vals)} cols")

    grid = df.pivot_table(index="y", columns="x", values="z", aggfunc="mean")
    grid = grid.reindex(index=y_vals, columns=x_vals)

    arr = grid.to_numpy(dtype="float32")
    arr = np.where(np.isnan(arr), nodata, arr)

    # north-up raster
    arr = np.flipud(arr)

    west = x_vals.min() - dx / 2
    north = y_vals.max() + dy / 2
    transform = from_origin(west, north, dx, dy)

    with rasterio.open(
        out_tif,
        "w",
        driver="GTiff",
        height=arr.shape[0],
        width=arr.shape[1],
        count=1,
        dtype="float32",
        crs=crs,
        transform=transform,
        nodata=nodata,
    ) as dst:
        dst.write(arr, 1)

    return out_tif
```

[13]:
```python
RASTER_OUT_DIR = DEPTH_GRID_DIR / "rasters_from_xyz"
RASTER_OUT_DIR.mkdir(parents=True, exist_ok=True)

out_tif = RASTER_OUT_DIR / "NSshublik_xy.tif"
xyz_to_raster_xy(test_df, out_tif, crs=None)

print("Wrote:", out_tif)
```

```
Grid size will be 208 rows x 435 cols
Wrote: /home/jovyan/notebooks/raw_data/north_slope_depth_grids/rasters_from_xyz/
NSshublik_xy.tif
```

[ ]:
```python
xyz_files = sorted(DEPTH_GRID_DIR.glob("*.XYZ")) + sorted(DEPTH_GRID_DIR.
 ↪glob("*.xyz"))
```

```python
raster_outputs = []

for xyz_path in xyz_files:
    try:
        df = read_xyz_file(xyz_path)
        out_tif = RASTER_OUT_DIR / f"{xyz_path.stem}.tif"
        xyz_to_raster_lonlat(df, out_tif)
        raster_outputs.append(out_tif)
        print(f"OK   -> {xyz_path.name} -> {out_tif.name}")
    except Exception as e:
        print(f"FAIL -> {xyz_path.name}: {e}")

print(f"\nCreated {len(raster_outputs)} rasters.")
```

```
/tmp/ipykernel_224/1769764110.py:16: PerformanceWarning: The following operation
may generate 4079985728 cells in the resulting pandas object.
  grid = df.pivot_table(index="lat", columns="lon", values="z", aggfunc="mean")
```

[14]:
```python
xyz_files = sorted(DEPTH_GRID_DIR.glob("*.XYZ")) + sorted(DEPTH_GRID_DIR.
 ↪glob("*.xyz"))

raster_outputs = []

for xyz_path in xyz_files:
    try:
        df = read_xyz_file(xyz_path)
        out_tif = RASTER_OUT_DIR / f"{xyz_path.stem}_xy.tif"
        xyz_to_raster_xy(df, out_tif, crs=None)
        raster_outputs.append(out_tif)
        print(f"OK   -> {xyz_path.name} -> {out_tif.name}")
    except Exception as e:
        print(f"FAIL -> {xyz_path.name}: {e}")

print(f"\nCreated {len(raster_outputs)} rasters.")
```

```
Grid size will be 208 rows x 435 cols
OK   -> NSLCU-Shublik.XYZ -> NSLCU-Shublik_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NSLCU.XYZ -> NSLCU_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NSbasement.XYZ -> NSbasement_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NSshublik-basement.XYZ -> NSshublik-basement_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NSshublik.XYZ -> NSshublik_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NStopo-LCU.XYZ -> NStopo-LCU_xy.tif
Grid size will be 208 rows x 435 cols
```

```
OK   -> NStopo-basement.XYZ -> NStopo-basement_xy.tif
Grid size will be 208 rows x 435 cols
OK   -> NStopo.XYZ -> NStopo_xy.tif

Created 8 rasters.
```

[ ]:
```python
from pathlib import Path
import subprocess
import sys

BASE_DIR = Path.cwd()
export_dir = BASE_DIR / "notebook_exports_pdf"
export_dir.mkdir(exist_ok=True)

#  find ALL notebooks in all subdirectories
notebooks = list(BASE_DIR.rglob("*.ipynb"))

print(f"Found {len(notebooks)} notebooks\n")

for nb_path in notebooks:
    print(f"Exporting {nb_path.name} -> PDF")

    cmd = [
        sys.executable,
        "-m",
        "jupyter",
        "nbconvert",
        "--to",
        "pdf",
        str(nb_path),
        "--output-dir",
        str(export_dir),
    ]

    result = subprocess.run(cmd, capture_output=True, text=True)

    if result.returncode == 0:
        print(f"DONE: {nb_path.name}")
    else:
        print(f"FAILED: {nb_path.name}")
        print(result.stderr)

print(f"\nFinished. Files are in:\n{export_dir}")
```

```
Found 2 notebooks

Exporting GISviz.ipynb -> PDF
```

```
[ ]:
```